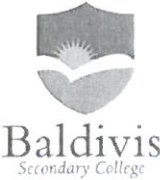
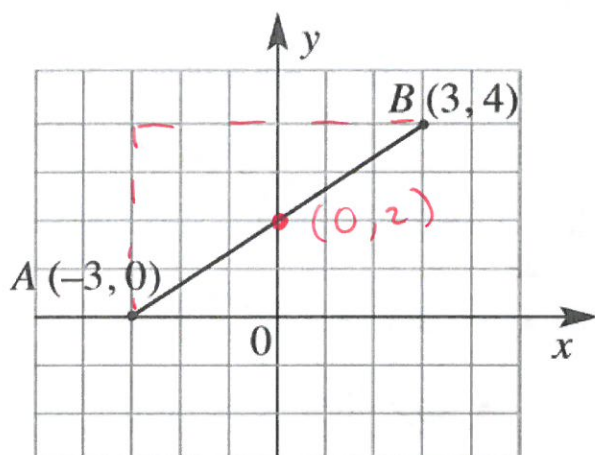


Name:	<u>Answers</u>	
Teacher:		
	Year 9 Mathematics Mini test Topic – Linear Equations	SCORE: /29 28
	<u>Full working out MUST be shown to get full marks for each question.</u>	
Total Time:	30 minutes	
Weighting:	7 %	
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.	

1. [5 marks] Vertical and horizontal distances
Consider the following graph



- a) What is the vertical distance between A and B?

$$✓ 4 - 0 = 4 \text{ units}$$

- b) What is the horizontal distance between A and B?

$$✓ 3 - (-3) = 6 \text{ units}$$

- c) What is the distance between A and B?

$$✓ \sqrt{(4)^2 + (6)^2} = \sqrt{52}$$

$$✓ \approx 7.21 \text{ units}$$

- d) What is the coordinate of the midpoint between A and B

$$✓ (0, 2)$$

2. [6 marks]

A line connects the two points $(-3, 5)$ and $(1, -3)$

- a) Find the distance between the two points

$$✓ \sqrt{(4)^2 + (8)^2} = \sqrt{16 + 64} = \sqrt{80} \approx 8.94 \text{ units}$$

- b) Find the mid-point

$$✓ x = \frac{-3 + 1}{2} = -1 \quad ; \quad y = \frac{5 + (-3)}{2} = 1 \quad \therefore (-1, 1)$$

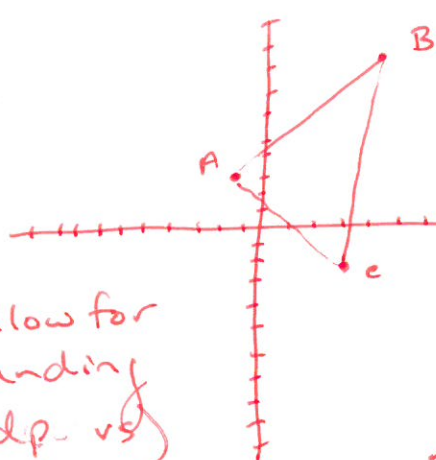
- c) What is the gradient of the line connecting the two points?

$$✓ \frac{y_2 - y_1}{x_2 - x_1} = \frac{-3 - 5}{1 - (-3)} = \frac{-8}{4} = -2$$

3. [4 marks]

A B C

A triangle is formed using the points $(-1, 3)$, $(4, 8)$ and $(3, -2)$ what is the perimeter of triangle?



$$\text{Per} = AB + BC + AC$$

$$\checkmark AB = \sqrt{(5)^2 + (5)^2} = \sqrt{50} \approx 7.07$$

$$\checkmark BC = \sqrt{(-1)^2 + (10)^2} = \sqrt{101} \approx 10.05$$

$$\checkmark AC = \sqrt{(4)^2 + (5)^2} = \sqrt{41} \approx 6.4$$

$$\checkmark \text{Per} = 7.07 + 10.05 + 6.4 \approx 23.52$$

* Allow for rounding
(1 dp. vs)
2 dp

4. [6 marks]

Sketch the following graphs on the set of axes (Label your graphs)

a) $y = x + 2$

b) $y = -2x + 3$

c) $2x + y = 10$

a)

x	0	1	2
y	2	3	4

 ✓

b)

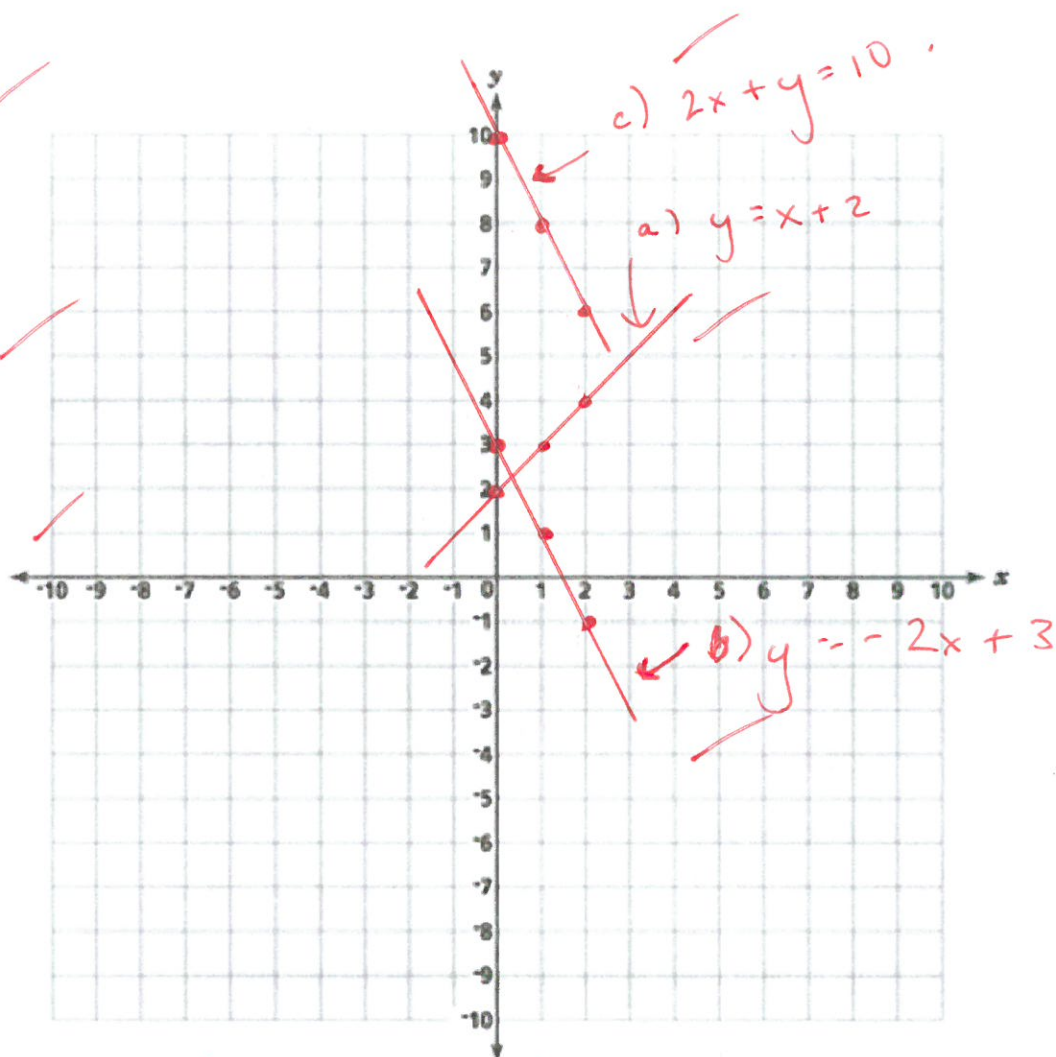
x	0	1	2
y	3	1	-1

 ✓

c)

x	0	1	2
y	10	8	6

 ✓



2 marks per question: 1 for table
1 for graphed line.

5. [4 marks]

Determine the rule for the following in the form $y = mx + c$

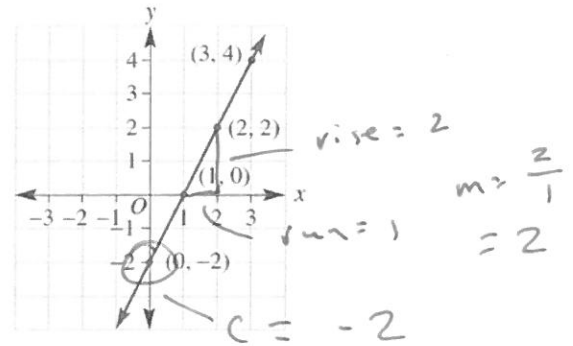
a)

x	0	1	2	3	4
y	3	5	7	9	11

$\underbrace{\quad\quad}_{+2} \quad \underbrace{\quad\quad}_{+2}$

$$y = 2x + 3$$

b)

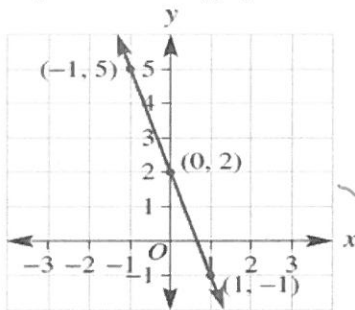


$$y = 2x - 2$$

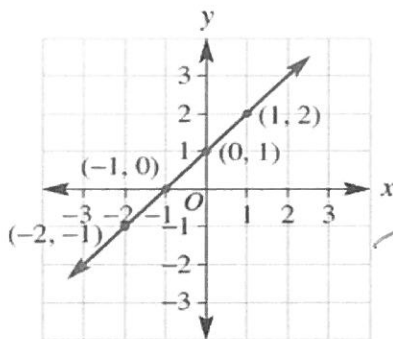
6. [4 marks]

Match the equation to the graph

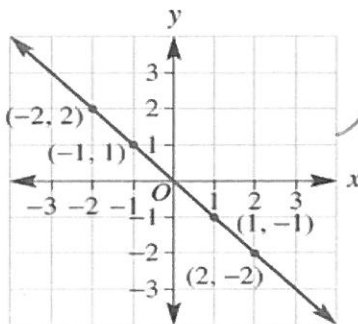
(a)



(b)



(c)



• $x + y = 0$ (1)

• $y - x = 1$ (2)

• $y = -3x + 2$ (3)

a : 3

b : 2

c : 1

End of Test

